Applications of Derivatives

File 2 of 3 · Last-Minute Revision Notes

Written for the night before the exam. Each topic: the idea in two lines, the formula, one easy, one medium and one hard worked example. Then a 'which method?' guide, the traps that cost marks, and a final checklist. Self-contained — you can revise from this alone. Questions marked PYQ appeared in Maharashtra HSC papers between Mar 2024 and Jun 2026.

Prepared for Ammaar Khatri • Source: Balbharati textbook (2020, reprint 2022), pp. 65–94; board papers Mar 2024 – Jun 2026

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1 The whole chapter in one idea

$f'(a)$ means three things at once. Every question in this chapter uses one of them:

$f'(a)$ as ...	Used for	One-line method
Slope	tangent, normal, Rolle, LMVT	slope $= \left(\frac{dy}{dx}\right)$ at the point
Rate	related rates, velocity, acceleration	differentiate the formula w.r.t. t
Direction / flatness	approximation, increasing–decreasing, max–min	sign of f'; zero of f'; sign of f''

Exam weight (observed in your six papers, not official): about **9 marks offered** every paper — usually one MCQ/VSA, one or two 2-markers, one or two 3-markers; a 4-mark optimisation question appeared in Mar 2024. Every question came straight from the textbook's solved examples or exercises.

2 Formula sheet

Memorise these

Tangent at (x_1, y_1)	$y - y_1 = m(x - x_1),\ m = \left(\frac{dy}{dx}\right)_{(x_1,y_1)}$
Normal	$y - y_1 = -\frac{1}{m}(x - x_1)$
Slope of $ax + by + c = 0$	$-\frac{a}{b}$
Parallel / perpendicular	$m_1 = m_2$ / $m_1 m_2 = -1$
Horizontal / vertical tangent	$\frac{dy}{dx} = 0$ / $\frac{dx}{dy} = 0$
Velocity, acceleration, jerk	$v = \frac{ds}{dt},\ a = \frac{d^2s}{dt^2},\ j = \frac{d^3s}{dt^3}$
Approximation	$f(a+h) \approx f(a) + h\,f'(a)$
Rolle	cont. $[a,b]$ + diff. (a,b) + $f(a) = f(b) \Rightarrow$ $f'(c) = 0, c \in (a,b)$
LMVT	cont. $[a,b]$ + diff. $(a,b) \Rightarrow f'(c) = \frac{f(b) - f(a)}{b - a}$
Increasing / decreasing	$f'(x) > 0$ / $f'(x) < 0$
Maximum at c	$f'(c) = 0,\ f''(c) < 0$
Minimum at c	$f'(c) = 0,\ f''(c) > 0$
If $f''(c) = 0$	test fails — check sign change of f' around c

Geometry formulas for rate and optimisation problems: circle $A = \pi r^2$; sphere $V = \frac{4}{3}\pi r^3$, $S = 4\pi r^2$; cylinder $V = \pi r^2 h$; cone $V = \frac{1}{3}\pi r^2 h$; cube $V = x^3$; triangle $\frac{1}{2}bc\sin A$; equilateral $\frac{\sqrt{3}}{4}a^2$.

3 Which method? (recognition guide)

If the question says ...	Do this
"equation of tangent / normal at ..."	find $\frac{dy}{dx}$, substitute point, point–slope form
"... at $\theta = ...$" or "$x = f(t),\ y = g(t)$"	parametric derivative **and** find the point (x, y)
"points where tangent is parallel to / perpendicular to ..."	equate $\frac{dy}{dx}$ to the required slope, solve for x, then y
"touches the curve at ... find a, b"	two equations: point lies on curve; slopes equal
"rate at which ... is increasing / how fast"	related rates: formula $\to \frac{d}{dt} \to$ substitute
"displacement $s = ...$"	$v = s'$, $a = s''$
"approximate value of ..."	$f(a+h) \approx f(a) + hf'(a)$ with a nice a
"verify Rolle's theorem" / "conditions satisfied?"	continuity, differentiability, $f(a) = f(b)$, then $f'(c) = 0$, $c \in (a, b)$
"verify LMVT" / "find c"	$f'(c) = \frac{f(b) - f(a)}{b - a}$, check $c \in (a, b)$
"test whether increasing or decreasing" / "find values of x for which ..."	sign of $f'(x)$ (factorise, sign table)
"maximum and minimum values of $f(x)$"	$f' = 0$, then f'' test
"dimensions ... maximum / minimum ..." (word problem)	constraint $\to$ objective in one variable $\to f' = 0$ $\to f''$ test

4 Tangents and normals

Idea. The tangent's slope is $\frac{dy}{dx}$ at the point; the normal is perpendicular to it.

EASY — explicit curve

Tangent and normal to $y = x^2 - 4x + 5$ at $(3, 2)$.
$\frac{dy}{dx} = 2x - 4$; at $x = 3$, $m = 2$; normal slope $-\frac{1}{2}$.
Tangent: $y - 2 = 2(x - 3) \Rightarrow 2x - y - 4 = 0$.
Normal: $y - 2 = -\frac{1}{2}(x - 3) \Rightarrow x + 2y - 7 = 0$.

MEDIUM — implicit curve

Tangent and normal to $x^2 + xy + y^2 = 7$ at $(1, 2)$.
Check: $1 + 2 + 4 = 7$ ✓. Differentiate: $2x + \left(x\frac{dy}{dx} + y\right) + 2y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{2x + y}{x + 2y}$.
At $(1, 2)$: $m = -\frac{4}{5}$; normal slope $\frac{5}{4}$.
Tangent: $y - 2 = -\frac{4}{5}(x - 1) \Rightarrow 4x + 5y - 14 = 0$.
Normal: $y - 2 = \frac{5}{4}(x - 1) \Rightarrow 5x - 4y + 3 = 0$.

HARD — parametric (the step everyone forgets is finding the point)

Tangent and normal to $x = a\cos^3\theta$, $y = a\sin^3\theta$ at $\theta = \frac{\pi}{4}$.
$\frac{dx}{d\theta} = -3a\cos^2\theta\sin\theta$, $\frac{dy}{d\theta} = 3a\sin^2\theta\cos\theta$, so $\frac{dy}{dx} = -\tan\theta$; at $\frac{\pi}{4}$: $m = -1$, normal slope 1.
Point: $x = y = a\left(\frac{1}{\sqrt{2}}\right)^3 = \frac{a}{2\sqrt{2}}$.
Tangent: $y - \frac{a}{2\sqrt{2}} = -\left(x - \frac{a}{2\sqrt{2}}\right) \Rightarrow x + y = \frac{a}{\sqrt{2}}$.
Normal: $y - \frac{a}{2\sqrt{2}} = x - \frac{a}{2\sqrt{2}} \Rightarrow x - y = 0$.

5 Parallel, perpendicular, "touches", finding points

Idea. Convert the geometric condition into an equation for $\frac{dy}{dx}$, solve for x, then get y **from the curve**.

EASY — [PYQ Jul 2024 Q9, Jul 2025 Q10]

Points on $y = x^3 - 6x^2 + x + 3$ where the tangent is parallel to $y = x + 5$.
$3x^2 - 12x + 1 = 1 \Rightarrow 3x(x-4) = 0 \Rightarrow x = 0, 4$. Points $(0, 3)$, $(4, -25)$.

MEDIUM — perpendicular

Points on $y = x^3 - 3x$ where the tangent is perpendicular to $x + 9y = 1$.
Slope of line $= -\frac{1}{9}$, so required slope $= 9$: $3x^2 - 3 = 9 \Rightarrow x = \pm 2$.
Points $(2, 2)$ and $(-2, -2)$.

HARD — tangent meets the curve again (Misc. I Q9)

The tangent at $(1, 1)$ to $y^2 = x(2-x)^2$ meets the curve again at P. Find P.
$2y\frac{dy}{dx} = (2-x)^2 - 2x(2-x)$; at $(1,1)$: $2\frac{dy}{dx} = 1 - 2 \Rightarrow m = -\frac{1}{2}$.
Tangent: $y - 1 = -\frac{1}{2}(x-1) \Rightarrow x + 2y = 3$, i.e. $y = \frac{3-x}{2}$.
Substitute: $\frac{(3-x)^2}{4} = x(2-x)^2 \Rightarrow 4x^3 - 17x^2 + 22x - 9 = 0 \Rightarrow (x-1)^2(4x-9) = 0$.
$x = 1$ is the point of tangency (double root), so $x = \frac{9}{4}$, $y = \frac{3}{8}$. $P = \left(\frac{9}{4}, \frac{3}{8}\right)$ — option (D).

Why a **double** root at $x = 1$: a tangent touches the curve, so the intersection there counts twice. This is a useful check.

6 Related rates

Idea. Formula linking the quantities → differentiate w.r.t. t → substitute the instant **after** differentiating. State units and "increasing/decreasing".

EASY — cube

Edge of a cube increases at 3 cm/s. Rate of increase of volume when edge = 10 cm?
$V = x^3 \Rightarrow \dfrac{dV}{dt} = 3x^2\dfrac{dx}{dt} = 3(100)(3) = 900$ cm³/s.

MEDIUM — sliding ladder (Misc. I Q3)

A 5 m ladder; foot pulled away at 1.5 m/s. Rate at which the top descends when the foot is 4 m from the wall?
$x^2 + y^2 = 25 \Rightarrow x\dfrac{dx}{dt} + y\dfrac{dy}{dt} = 0$. At $x = 4$: $y = 3$. $\dfrac{dy}{dt} = -\dfrac{4(1.5)}{3} = -2$.
The top slides down at 2 m/s — option (B).

HARD — water into an inverted cone

Water flows at 5 cm³/s into an inverted cone of height 10 cm and base radius 5 cm. How fast is the level rising when depth is 4 cm?
Similar triangles: $\dfrac{r}{h} = \dfrac{5}{10} \Rightarrow r = \dfrac{h}{2}$ (fixed ratio — substitute before differentiating).
$V = \dfrac{1}{3}\pi\left(\dfrac{h}{2}\right)^2 h = \dfrac{\pi h^3}{12} \Rightarrow \dfrac{dV}{dt} = \dfrac{\pi h^2}{4}\dfrac{dh}{dt}$.
$5 = \dfrac{\pi(16)}{4}\dfrac{dh}{dt} \Rightarrow \dfrac{dh}{dt} = \dfrac{5}{4\pi}$ cm/s.

Other high-frequency rate questions: circular wave 3 cm/s at $r = 8 \Rightarrow \dfrac{dA}{dt} = 48\pi$ cm²/s **[PYQ Feb 2026]**; water 27 m³/s into cylinder radius 3 $\Rightarrow \dfrac{dh}{dt} = \dfrac{3}{\pi}$ m/s **[PYQ Jul 2025]**; lamp-post (similar triangles) gives a rate independent of position.

7 Velocity and acceleration

EASY

$s = t^3 - 3t^2 + 5$. Velocity and acceleration at $t = 2$.
$v = 3t^2 - 6t = 0$, $a = 6t - 6 = 6$. (Momentarily at rest at $t = 2$.)

MEDIUM — [PYQ Jul 2024 Q11; Feb 2025 Q22]

$s = 2t^3 - 5t^2 + 4t - 3$. Velocity and displacement when acceleration is 14 ft/s².
$v = 6t^2 - 10t + 4$, $a = 12t - 10 = 14 \Rightarrow t = 2$.
$v(2) = 8$ ft/s, $s(2) = 1$ ft.

HARD — when at rest, and jerk

$s = t^3 - 6t^2 + 9t + 1$. When is the particle at rest? Find acceleration then, and the jerk.
$v = 3t^2 - 12t + 9 = 3(t-1)(t-3) = 0 \Rightarrow t = 1, 3$.
$a = 6t - 12$: $a(1) = -6$, $a(3) = 6$. Jerk $j = \dfrac{da}{dt} = 6$ (constant).

8 Approximations

Idea. Walk along the tangent: $f(a+h) \approx f(a) + hf'(a)$. Pick a where everything is exact. **Trig: h in radians.**

EASY

$\sqrt{36.3}$: $f = \sqrt{x}$, $a = 36$, $h = 0.3$, $f'(36) = \dfrac{1}{12}$.
$\sqrt{36.3} \approx 6 + \dfrac{0.3}{12} = 6.025$.

MEDIUM — [PYQ Mar 2024 Q22]

$\tan^{-1}(1.002)$, $\pi = 3.1416$: $a = 1$, $h = 0.002$, $f(1) = \dfrac{\pi}{4} = 0.7854$, $f'(1) = \dfrac{1}{2}$.
$\tan^{-1}(1.002) \approx 0.7854 + 0.001 = 0.7864$.

HARD — degrees and minutes

$\cos(60°30')$, given $1° = 0.0175^c$, $\sqrt{3} = 1.732$.
$f = \cos x$, $f' = -\sin x$; $a = \dfrac{\pi}{3}$, $h = 30' = 0.00875^c$.
$\cos(60°30') \approx \dfrac{1}{2} - (0.00875)\dfrac{\sqrt{3}}{2} = 0.5 - (0.00875)(0.866) = 0.5 - 0.0075775 = 0.4924225$.

PYQ MCQs: $\tan(44°30') \approx 0.9825$ (Jul 2024); $x^3 - 3x + 5$ at $1.99 \approx 6.91$ (Feb 2026). Polynomial at 1.98: $x^3 + 5x^2 - 2x + 3 \approx 26.4$ (Jul 2024, 3 marks).

9 Rolle's theorem and LMVT

Idea. Rolle: equal heights at the ends ⇒ a flat tangent in between. LMVT: some tangent is parallel to the chord. Always **state** continuity on $[a,b]$ and differentiability on (a,b) with a reason, and **check** $c \in (a,b)$.

EASY — check conditions [PYQ Jul 2025 Q11]

$f(x) = x^2 - 4x + 3$ on $[1,3]$: polynomial ⇒ continuous on $[1,3]$, differentiable on $(1,3)$; $f(1) = 0 = f(3)$. Conditions satisfied. ($f'(c) = 2c - 4 = 0 \Rightarrow c = 2 \in (1,3)$.)

MEDIUM — verify LMVT [PYQ Feb 2026 Q22]

$f(x) = \log x$ on $[1, e]$: continuous and differentiable for $x > 0$. $f(1) = 0$, $f(e) = 1$.
$\frac{1}{c} = \frac{1-0}{e-1} \Rightarrow c = e - 1$; since $2 < e < 3$, $c \in (1, e)$. Verified.

HARD — exponential factor (Ex. 2.3 Q6)

$f(x) = x(x+3)e^{-x/2}$ on $[-3, 0]$. Find c of Rolle's theorem.
$f(-3) = 0 = f(0)$. Product rule:
$f'(x) = (2x+3)e^{-x/2} - \frac{1}{2}(x^2+3x)e^{-x/2} = -\frac{1}{2}e^{-x/2}(x^2 - x - 6) = -\frac{1}{2}e^{-x/2}(x-3)(x+2)$.
$e^{-x/2} \neq 0$, so $x = 3$ or -2. Only $c = -2 \in (-3, 0)$.

Not applicable when: corner ($|x|$), cusp ($x^{2/3}$ at 0), a piecewise joint with unequal one-sided derivatives, or $f(a) \neq f(b)$ (Rolle only). Write "**not applicable**", not "false".

10 Increasing and decreasing

Idea. $f' > 0 \Rightarrow$ increasing; $f' < 0 \Rightarrow$ decreasing. Factorise f', mark its zeros, use a sign table.

EASY

$f(x) = x^3 + 10x + 7$: $f'(x) = 3x^2 + 10 > 0$ for all x (since $3x^2 \geq 0$). Strictly increasing on $\mathbb{R}$.

MEDIUM — intervals

$f(x) = 2x^3 - 3x^2 - 12x + 6$. $f'(x) = 6(x^2 - x - 2) = 6(x-2)(x+1)$.

Interval	$x+1$	$x-2$	f'	f
$x < -1$	$-$	$-$	$+$	increasing
$-1 < x < 2$	$+$	$-$	$-$	decreasing
$x > 2$	$+$	$+$	$+$	increasing

HARD — $f(x) = x^x$, $x > 0$ (Misc. II Q10)

$\log f = x \log x \Rightarrow \frac{f'}{f} = \log x + 1 \Rightarrow f'(x) = x^x(1 + \log x)$.
$x^x > 0$, so sign of f' = sign of $1 + \log x$: positive when $\log x > -1$, i.e. $x > \frac{1}{e}$.
Increasing for $x > \frac{1}{e}$; decreasing for $0 < x < \frac{1}{e}$.

PYQs: $x - \frac{1}{x}$ is increasing since $f' = 1 + \frac{1}{x^2} > 0$ (Jun 2026); $x^3 + 6x^2 + 12x - 7$ is increasing since $f' = 3(x+2)^2 \geq 0$ (Feb 2026); "condition for strictly increasing": $f'(x) > 0$ (Feb 2025, 1 mark).

11 Maxima and minima of a function

Idea. Stationary points from $f' = 0$; classify with f'' (negative $\Rightarrow$ max, positive $\Rightarrow$ min). Then compute f at those points.

EASY

$f(x) = x^2 - 6x + 10$: $f' = 2x - 6 = 0 \Rightarrow x = 3$; $f'' = 2 > 0$; minimum $f(3) = 1$.

MEDIUM — textbook Ex. 1 (p. 85)

$f(x) = x^3 - 3x^2 - 24x + 5$: $f' = 3(x+2)(x-4)$, $f'' = 6x - 6$.
$f''(-2) = -18 < 0$: max $f(-2) = 33$. $f''(4) = 18 > 0$: min $f(4) = -75$.

HARD — $f(x) = \dfrac{\log x}{x}$

$$f'(x) = \frac{x \cdot \frac{1}{x} - \log x}{x^2} = \frac{1 - \log x}{x^2} = 0 \Rightarrow x = e.$$

$$f''(x) = \frac{-\frac{1}{x} \cdot x^2 - (1 - \log x)2x}{x^4} = \frac{2\log x - 3}{x^3}; \; f''(e) = -\frac{1}{e^3} < 0.$$

Maximum value $f(e) = \dfrac{1}{e}$.

12 Optimisation word problems (the 4-marker)

Template: variables → constraint → objective in **one** variable → $f' = 0$ → reject impossible roots → f'' test → answer with units.

EASY — [PYQ Feb 2025 Q9 type]

Two positive numbers with sum 24 and maximum product.
$P = x(24 - x)$, $P' = 24 - 2x = 0 \Rightarrow x = 12$, $P'' = -2 < 0$. Numbers 12, 12; product 144.

MEDIUM — [PYQ Mar 2024 Q32, 4 marks]

Open box, square base, surface area 147 cm². Dimensions for maximum volume.

Base side x, height h: $x^2 + 4xh = 147 \Rightarrow h = \dfrac{147 - x^2}{4x}$.

$$V = x^2 h = \frac{147x - x^3}{4}; \; V' = \frac{147 - 3x^2}{4} = 0 \Rightarrow x = 7; \; V'' = -\frac{3x}{2} < 0 \Rightarrow \text{max}.$$

$h = \dfrac{98}{28} = 3.5$. Box: 7 cm × 7 cm × 3.5 cm.

HARD — closed cylinder of given surface area, maximum volume (Misc. II Q14)

Show height = diameter.

$S = 2\pi r^2 + 2\pi r h$ (constant) $\Rightarrow h = \dfrac{S - 2\pi r^2}{2\pi r}$.

$$V = \pi r^2 h = \frac{r(S - 2\pi r^2)}{2} = \frac{Sr - 2\pi r^3}{2}.$$
$V' = \dfrac{S - 6\pi r^2}{2} = 0 \Rightarrow S = 6\pi r^2$; $V'' = -6\pi r < 0 \Rightarrow$ max.
Then $2\pi r^2 + 2\pi rh = 6\pi r^2 \Rightarrow h = 2r$ = diameter. ■

Classic results worth knowing (they let you *check* answers fast): rectangle of given perimeter with max area is a **square**; open square-based box of max volume has $h = \dfrac{\text{side}}{2}$; closed cylinder (given volume, least surface, or given surface, max volume) has $h = 2r$; cylinder in a sphere of radius R: $h = \dfrac{2R}{\sqrt{3}}$; cone in a sphere: altitude $\dfrac{4R}{3}$; cylinder in a cone: $h = \dfrac{H}{3}$.

13 Frequently confused pairs

Don't confuse	Difference
Slope of normal $-\dfrac{1}{m}$ vs $-m$	perpendicular $\Rightarrow$ product $= -1 \Rightarrow -\dfrac{1}{m}$
Rolle vs LMVT	Rolle needs $f(a) = f(b)$ and gives $f'(c) = 0$; LMVT needs no equal ends, gives slope of chord
$[a, b]$ vs (a, b)	continuity on **closed**, differentiability and c on **open**
Increasing ($f' \geq 0$) vs strictly increasing ($f' > 0$)	x^3 is strictly increasing with $f'(0) = 0$
Local max vs greatest value	local max is highest only *nearby*
$f''(c) = 0$	the test fails — it does **not** mean "no extremum"
Speed vs velocity	speed $= \lvert v \rvert$
h in degrees vs radians	always radians: $1^\circ = 0.0175^c$, $30' = 0.00875^c$

14 Traps that cost marks

Top 10 mark-losers

1. Substituting the point (or a changing quantity) **before** differentiating.
2. Not finding (x, y) in parametric tangent questions.
3. Normal slope written as $-m$ or $\dfrac{1}{m}$.
4. Answer not in $ax + by + c = 0$ form with fractions cleared.
5. Rate answer without units or without saying increasing/decreasing.
6. Approximation with h in degrees; writing "$=$" instead of "$\approx$".
7. Rolle/LMVT without stating continuity and differentiability (with a reason).
8. Keeping a value of c outside (a, b).
9. Optimisation without the second-derivative test, or without rejecting impossible roots (like $x = 9$ in the 18 cm box).
10. Giving the stationary x but not the maximum/minimum **value** when the value is asked.

Memory tricks

- **Smile = min.** $f'' > 0$ is a cup $(\smile)$ — it holds water at the bottom: minimum.
- **Rolle = Return.** Leave home and return $\Rightarrow$ at some instant your velocity was zero.
- **LMVT = Toll camera.** Average speed over a stretch is achieved at some instant.
- **Approximation = Tangent walk.** New value = old value + step × slope.
- **Related rates = Formula first, numbers last.**

15 Rapid-revision checklist

Tick each only if you can do it without notes.

- ☐ Write the tangent and normal at a point for explicit, implicit and parametric curves
- ☐ Find points where the tangent is parallel/perpendicular to a given line
- ☐ Find constants when a line touches a curve at a given point
- ☐ Solve a related-rates question (circle, sphere, cylinder, cone, ladder, shadow) with units
- ☐ Get v, a, j from $s(t)$; find when the particle is at rest
- ☐ Approximate roots, powers, log, e^x, trig and inverse trig values (radians!)
- ☐ State Rolle and LMVT exactly, with geometric meaning
- ☐ Verify Rolle/LMVT, and explain why a function fails the conditions
- ☐ Find intervals of increase/decrease with a sign table
- ☐ Find local max/min values using f''
- ☐ Set up and solve: wire-rectangle, open box, box from a sheet, cylinder, cone, window problems
- ☐ Recall: square for max area, $h = \dfrac{\text{side}}{2}$ for open box, $h = 2r$ for cylinder